

A Reexamination of the Classical PKN Model of Hydraulic Fracture

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Abstract This article reexamines the classical PKN model of hydraulic fracture Perkins and Kern (J. Pet. Tech. Trans. AIME, 222:937–949 (1961)) and Nordgren (J. Pet. Tech. 253:306–314 (1972)) using novel approaches, which have recently been developed to tackle this class of problems that are characterized by a moving boundary and strong non-linearities in the governing equations. First, we demonstrate, using scaling arguments only, that a PKN hydraulic fracture has two limiting time asymptotic behaviors: storage-dominated at small time, and leak-off-dominated at large time. Next, we investigate the multiscale nature of the tip asymptotics and its implication for the construction of a robust and efficient numerical algorithm. In particular, we show that in the storage-dominated regime the tip aperture w behaves according to $w \sim x^{1/3}$ (where x is the distance from the tip), and in the leak-off-dominated regime according to $w \sim x^{3/8}$. However, the solution in the leak-off-dominated regime has a boundary layer structure at the tip, with $w \sim x^{3/8}$ acting as an intermediate asymptote that matches an inner solution with a $w \sim x^{1/3}$ asymptote to the outer (global) solution. Finally, we describe an efficient numerical algorithm with the multiscale tip asymptotics embedded in the tip element, which is used to compute the transient solution that connects the small- and large-time asymptotes.

Keywords Hydraulic fracture · PKN model

1 Introduction

This article deals with the classical PKN (Perkins–Kern–Nordgren) model of hydraulic fracture (Perkins and Kern 1961; Nordgren 1972). This model is widely used in the Oil and Gas Industry to assist in the design of hydraulic fracturing treatments that are engineered

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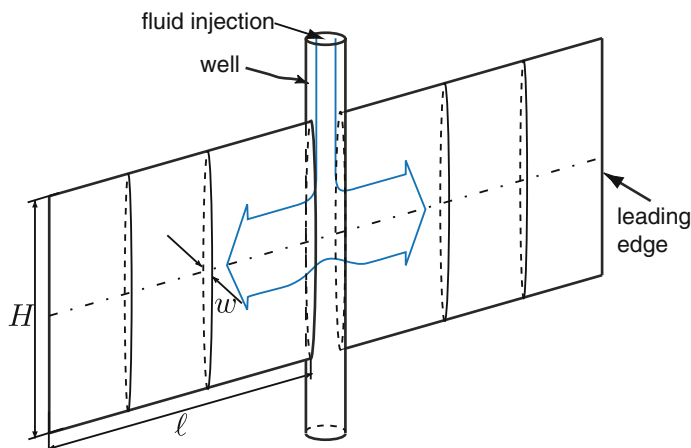


Fig. 1 PKN Hydraulic fracture

to enhance the recovery of hydrocarbons from underground reservoirs (Economides and Nolte 2000; Mendelsohn 1984; Valkó and Economides 1995). The PKN model is applicable for conditions when the vertical hydraulic fracture remains confined within the horizontal permeable layer, on account of a sufficiently high contrast in horizontal stress between the reservoir layer and the adjacent impermeable layers. In other words, the PKN model is based on the assumption that the hydraulic fracture propagates laterally with a constant height H corresponding to the thickness of the reservoir, see Fig. 1. Given these conditions, and provided that the fracture half-length ℓ is much larger than H (the fracture is assumed to consist of two symmetric wings), propagation of the hydraulic fracture can be simulated using the one-dimensional PKN model. In this model, the field quantities are averaged over the height of the fracture, and the elastic relationship between the fracture aperture and the net pressure is assumed to be local. Since, the fracture half-length ℓ is a function of time t , the PKN consistency condition that $\ell/H \gg 1$ places obvious restriction on the minimum time at which the model becomes applicable.

Surprisingly, only a handful of papers deal with the mathematical formulation and the construction of solutions for the PKN fracture model, despite its importance for the design of hydraulic fracturing treatments when the fracture remains confined to the reservoir (Perkins and Kern 1961; Nordgren 1972; Kemp 1990; Gordeyev and Zazovsky 1992). These contributions are now briefly reviewed. The original formulation of the PKN model is credited to Perkins and Kern (1961); however, the storage effect associated with the change of aperture taking place when the fracture propagates is ignored in this landmark contribution, as well as the leak-off of fracturing fluid into the reservoir. Nordgren (1972) corrected the model to account for the storage effect and also included a leak-off term in the equations that behave according to Carter law (1957), i.e., the leak-off rate is inversely proportional to the square root of the time elapsed, since first exposure to the fracturing fluid. Nordgren (1972) also provided numerical solutions of the non-linear partial differential equation governing this problem, as well as closed-form small- and large-time asymptotic solutions. Some ambiguity in the boundary condition at the moving front in Nordgren's article is discussed by Kemp (1990), who provided the correct asymptotic form of the fracture width in the tip region in the case of zero leak-off. This asymptote is constructed on the recognition that the average fluid velocity in the tip region is equal to the tip velocity in the absence of leak-off. However,

the nature of the boundary conditions at the tip when there is leak-off remains ambiguous, due in part, as explained in this article, to the presence of a boundary layer at the tip. Construction of self-similar solutions for the PKN model when the leak-off is not restricted by Carter's law is presented by Gordeyev and Zazovsky (1992), who derived solutions for a constant pressure at the fracture inlet, by neglecting the pore pressure gradient along the crack. Approximated solutions for the case where the injection rate is prescribed at the inlet are proposed by Economides et al. (2007).

In fact, the formulation of the PKN fracture model gives rise to a challenging mathematical problem, in contrast to the simplicity of the physical problem that it aims to capture. Indeed, this problem is governed by a strongly non-linear partial differential equation and is characterized by the presence of a moving boundary and by a degeneracy of the governing equation near the moving tip due to the vanishing of the fracture aperture. Furthermore, leak-off of fluid in the reservoir is responsible for a history-dependent forcing function in the non-linear PDE. In view of the complexity of the underlying mathematical model, there is merit in analyzing the structure of the equations, in particular with the goal of establishing the asymptotics of the solution near the moving tip of the fracture, and also of constructing the small- and large-time asymptotic solutions of this evolution problem. Knowledge of the asymptotics of this problem, combined with scaling of the equations can indeed have a significant impact on the efficiency, accuracy, and robustness of the numerical technique needed to solve the transient problem.

The goal of the present article is to reexamine the classical PKN fracture model from the dual perspective of scaling and tip asymptotics, which has buttressed some recent study in hydraulic fracturing modeling, see for example Desroches et al. (1994), Garagash and Detournay (2000), Adachi and Detournay (2002), Detournay (2004), and Mitchell et al. (2007b). Indeed, scaling is directly linked to the search of self-similar solutions of the system of equations. These solutions generally correspond to particular asymptotics of the problem (Barenblatt 1996), and more precisely to small- and large-time asymptotes in problems characterized by one time scale, as the one of concern here. Moreover, establishment of the tip asymptotics is of crucial importance for developing accurate numerical solution, in view of the degeneracy of the governing equation near the tip. These asymptotes can indeed be embedded in the tip element and naturally act as boundary conditions for the problem at large.

This article is organized as follows: a detailed description of the mathematical model is included in Sect. 2. Scaling and the existence of different propagation regimes are described in Sect. 3. A discussion on the near-tip behavior of the propagating fracture is given in Sect. 4. The asymptotic solutions for small and large time are derived in Sect. 5, and the numerical solution is presented in Sect. 6. Finally, the paper is concluded in Sect. 7.

2 Mathematical Model

2.1 Problem Definition

We consider the propagation of a PKN fracture driven by injection of an incompressible Newtonian fluid at constant volumetric flow Q_0 , from a fixed line source located at the center of the fracture. A far-field compressive stress of magnitude σ_0 acts perpendicular to the fracture. A system of coordinates (x, z) is defined in such a way that the x -axis coincides with the direction of crack propagation, with its origin at mid-height on the line source, see Fig. 2. We seek to find the evolution of the fracture half-length ℓ , as well as the average aperture w in a cross-section as a function of both position x and time t . This solution depends on five

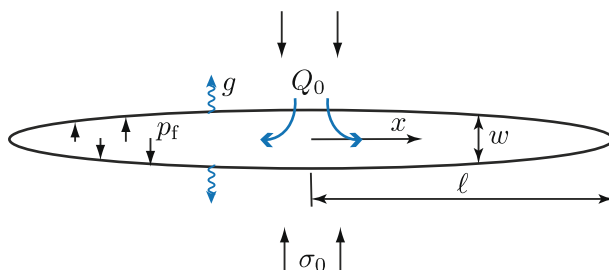


Fig. 2 Equivalent lumped one-dimensional model

additional parameters: the fracture height H , the injection flow rate Q_0 , Young's modulus E and Poisson's ratio ν of the rock, the dynamic viscosity μ of the fracturing fluid, and the leak-off coefficient C_1 , a property of both the rock and the fluid. However, we will use the following reduced parameters, which are introduced to avoid carrying numerical factors in the equations

$$\bar{E} = \frac{2}{\pi} \frac{E}{1 - \nu^2}, \quad \bar{\mu} = \pi^2 \mu, \quad C' = 2C_1. \quad (2.1)$$

The far-field stress σ_0 only enter the solution as a reference to the fluid pressure.

2.2 Governing Equations

The mass conservation law of the incompressible fracturing fluid can be written as (Nordgren 1972)

$$\frac{\partial q}{\partial x} + \frac{\partial w}{\partial t} + g = 0. \quad (2.2)$$

Here, $w(x, t)$ is the average fracture width, $q(x, t)$ is the average flow rate per unit height of the fracture, and $g(x, t)$ is the leak-off term also defined per unit fracture height. Thus, $w(x, t)H$, $q(x, t)H$, and $g(x, t)H$ represent, respectively, the vertical cross-sectional area of the fracture, the longitudinal flux, and the total leak-off rate at position x and at time t .

Using the plane strain approximation, the average crack aperture w is related to the local net pressure $p(x, t) = p_f - \sigma_0$, the difference between the fracturing fluid pressure p_f and the confining stress σ_0 (Nordgren 1972)

$$w = \frac{H}{\bar{E}} p. \quad (2.3)$$

It is assumed that the flow rate is related to the net pressure gradient by the classical solution for laminar flow of a Newtonian viscous fluid in a channel of elliptical cross-section (Nordgren 1972)

$$q = -\frac{w^3}{\bar{\mu}} \frac{\partial p}{\partial x}. \quad (2.4)$$

Substituting (2.3) into (2.4) yields an expression for the fluid flow in terms of the crack opening

$$q = -\frac{\bar{E}}{4\bar{\mu}H} \frac{\partial w^4}{\partial x}. \quad (2.5)$$

The fluid leak-off is treated as a one-dimensional process in accordance to Carter's model (1957)

$$g = \frac{C'}{\sqrt{t - t_0(x)}}, \quad (2.6)$$

where $t_0(x)$ is the time at which the crack arrived at a point of coordinate x . The usual assumptions behind this model are that (i) the fracturing fluid deposits a thin layer of relatively low permeability known as the *filter cake* on the inner faces of the fracture, with the deposition rate being proportional to the leak-off rate, and (ii) the filtrate has enough viscosity to fully displace the fluid already present in the rock pores.

The three governing Eqs. 2.2, 2.4, and 2.6 can be combined into one non-linear partial differential equation for the aperture $w(x, t)$ with a delay term

$$\frac{\partial w}{\partial t} - \frac{\bar{E}}{4\bar{\mu}H} \frac{\partial^2 w^4}{\partial x^2} + \frac{C'}{\sqrt{t - t_0(x)}} = 0. \quad (2.7)$$

For convenience, we will refer to (2.7) as the Reynolds equation.

The boundary conditions at the crack tip $x = \ell$ are

$$w = q = 0, \quad \text{at } x = \ell(t), \quad (2.8)$$

where q can be expressed in terms of the gradient of w^4 in view of (2.5). At the fluid injection point $x = 0$, the fluid flow rate is constant and equal to

$$q = \frac{Q_0}{2H}, \quad \text{at } x = 0. \quad (2.9)$$

The initial conditions are formally given by

$$\ell = w = q = 0, \quad \text{at } t = 0. \quad (2.10)$$

The system of Eqs. 2.7–2.10 governing the evolution of the fracture is closed and can thus be solved for the fracture length $\ell(t)$, and the field $w(x, t)$. Note that the boundary condition (2.9) at the fracture inlet can be replaced by the global continuity equation

$$\frac{Q_0 t}{2H} = \int_0^{\ell(t)} w dx + 2C' \int_0^{\ell(t)} \sqrt{t - t_0(x)} dx, \quad (2.11)$$

which is obtained by integrating the local balance Eq. 2.2 both in time and space and using (2.6) as well as the initial and boundary conditions.

3 Scaling

3.1 Preamble

The dependence of this problem on the variables x and t and on the parameters H , Q_0 , $\bar{E}$, $\bar{\mu}$, and C' can, in fact, be reduced to a dependence on the scaled spatial coordinate $\xi = x/\ell(t)$, and on a dimensionless evolution parameter $\mathcal{P}$. In other words, the solution is of the form $\mathcal{F} = \mathcal{F}(\xi, \mathcal{P})$. This evolution parameter $\mathcal{P}$ is either a dimensionless time τ or a certain power

law of τ depending on the scaling used, as discussed below. The dimensionless time τ is defined as

$$\tau = \frac{t}{t_{m\tilde{m}}}, \quad (3.1)$$

where $t_{m\tilde{m}}$ is the problem time scale associated with the fluid leak-off process, see discussion further below and in Appendix A

$$t_{m\tilde{m}} = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^2}{C'^5 H} \right)^{2/3}. \quad (3.2)$$

During fracture propagation, the injected volume—the left-hand side of the global continuity Eq. 2.11 splits into two components: (i) the fracture volume—the first term of the right-hand side of (2.11), and (ii) the volume of fluid that has leaked-off into the surrounding medium—the second term of the right-hand side of (2.11). In fact, the fracture evolves between two asymptotic regimes with the characteristic time $t_{m\tilde{m}}$: one in which fluid storage in the fracture dominates at small time (“storage-viscosity-dominated regime”); and one in which leak-off of fluid into surrounding medium dominates at large time (“leak-off-viscosity-dominated regime”). We will refer to the former as the “M-regime,” and to the latter as the “ $\tilde{M}$ -regime.”

3.2 Scaling Schemes

After identifying the M- and $\tilde{M}$ -regimes, we proceed to introduce two scaling schemes of the form (Detournay 2004)

$$\ell(t) = L(t) \gamma(\mathcal{P}), \quad w(x, t) = W(t) \Omega(\xi, \mathcal{P}). \quad (3.3)$$

Here, $W(t)$ and $L(t)$ are power law of time functions that represent (time-dependent) length scales expected to have the same order as the fracture opening and the fracture half-length, respectively, and Ω, γ are dimensionless crack opening and length. The variable $\xi = x/\ell$ ($0 \leq \xi \leq 1$) defines a stretching system of coordinates.

The details on how to devise these two scaling schemes, referred to as the M- and $\tilde{M}$ -scaling, can be found in Appendix A. We only summarize here the results, using subscript m ($\tilde{m}$) to denote quantities in the M- ($\tilde{M}$ -)scaling.

In the M-scaling, the quantities W, L , and $\mathcal{P}$ take the explicit forms

$$W_m = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^2}{H} \right)^{1/5} t^{1/5}, \quad L_m = \left(\frac{\bar{E}}{\bar{\mu}} \frac{Q_0^3}{H^4} \right)^{1/5} t^{4/5}, \quad C_m = C' \left(\frac{\bar{E}}{\bar{\mu}} \frac{H t^{3/2}}{Q_0^2} \right)^{1/5}, \quad (3.4)$$

and in the $\tilde{M}$ -scaling

$$W_{\tilde{m}} = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^2}{C' H} \right)^{1/4} t^{1/8}, \quad L_{\tilde{m}} = \frac{Q_0}{C' H} t^{1/2}, \quad S_{\tilde{m}} = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^2}{C'^5 H t^{3/2}} \right)^{1/4}. \quad (3.5)$$

Thus, the evolution parameter $\mathcal{P}$ is identified as the dimensionless leak-off coefficient $C_m = \tau^{3/10}$ in the M-scaling and as $S_{\tilde{m}} = \tau^{-3/8}$ in the $\tilde{M}$ -scaling.

In order to reduce the dependence on C_m and γ_m of the governing equations in the M-scaling to only a dependence on $\bar{C}_m = \gamma_m^{-2/3} C_m$, we introduce the quantity $\bar{\Omega}_m = \gamma_m^{-2/3} \Omega_m$.

Then the Reynolds Equation (2.7), the global fluid conservation (2.11), and the boundary condition (2.9) in this scaling are given by (see Appendix A for details of the derivation)

$$\frac{1}{2} = \gamma_m^{5/3} \int_0^1 \bar{\Omega}_m d\xi + 2\gamma_m^{5/3} \bar{C}_m \int_0^1 \sqrt{1 - \theta(\xi, \tau)} d\xi, \quad (3.6)$$

$$\left[\left(\frac{1}{5} + \frac{2}{3} \frac{\dot{\gamma}_m \tau}{\gamma_m} \right) \bar{\Omega}_m + \dot{\bar{\Omega}}_m \tau - \xi \left(\frac{\dot{\gamma}_m \tau}{\gamma_m} + \frac{4}{5} \right) \frac{\partial \bar{\Omega}_m}{\partial \xi} \right] + \frac{\bar{C}_m}{\sqrt{1 - \theta}} = \frac{1}{4} \frac{\partial^2 \bar{\Omega}_m^4}{\partial \xi^2}, \quad (3.7)$$

$$\left. \frac{\partial \bar{\Omega}_m^4}{\partial \xi} \right|_{\xi=0} = -2\gamma_m^{-5/3}. \quad (3.8)$$

Similarly, the solution in the $\tilde{M}$ -scaling is governed by the following scaled equations, in which we have introduced the substitution $\bar{\Omega}_{\tilde{m}} = \gamma_{\tilde{m}}^{-1/2} \Omega_{\tilde{m}}$ and the dimensionless quantity $\bar{S}_{\tilde{m}} = \gamma_{\tilde{m}}^{1/2} S_{\tilde{m}}$

$$\frac{1}{2} = \gamma_{\tilde{m}} \bar{S}_{\tilde{m}} \int_0^1 \bar{\Omega}_{\tilde{m}} d\xi + 2\gamma_{\tilde{m}} \int_0^1 \sqrt{1 - \theta(\xi, \tau)} d\xi, \quad (3.9)$$

$$\bar{S}_{\tilde{m}} \left[\left(\frac{1}{8} + \frac{1}{2} \frac{\dot{\gamma}_{\tilde{m}} \tau}{\gamma_{\tilde{m}}} \right) \bar{\Omega}_{\tilde{m}} + \dot{\bar{\Omega}}_{\tilde{m}} \tau - \xi \left(\frac{\dot{\gamma}_{\tilde{m}} \tau}{\gamma_{\tilde{m}}} + \frac{1}{2} \right) \frac{\partial \bar{\Omega}_{\tilde{m}}}{\partial \xi} \right] + \frac{1}{\sqrt{1 - \theta}} = \frac{1}{4} \frac{\partial^2 \bar{\Omega}_{\tilde{m}}^4}{\partial \xi^2}, \quad (3.10)$$

$$\left. \frac{\partial \bar{\Omega}_{\tilde{m}}^4}{\partial \xi} \right|_{\xi=0} = -2\gamma_{\tilde{m}}^{-1}. \quad (3.11)$$

In the above Eqs, $(\cdot)$ indicates the time derivative in the coordinate system (ξ, τ) and $\theta = t_0/t$ is the normalized arrival time.

Through scaling, it has thus been established that there is a universal solution of the form $\mathcal{F}(\xi, \tau)$ (or equivalently $\mathcal{F}_m(\xi, C_m)$ or $\mathcal{F}_{\tilde{m}}(\xi, S_{\tilde{m}})$) to the original problem, which can then readily be translated into a dimensional form for particular values of the physical parameters. It is an important result, which was already obtained by Nordgren (1972). Furthermore, the scaling analysis indicates the existence of two similarity solutions $\mathcal{F}_{m0}(\xi) = \mathcal{F}_m(\xi, 0)$ (the M-solution) and $\mathcal{F}_{\tilde{m}0}(\xi) = \mathcal{F}_{\tilde{m}}(\xi, 0)$ (the $\tilde{M}$ -solution), which can be determined by solving Eqs. 3.7–3.8 and 3.10–3.11, respectively. As discussed previously, the M-solution corresponds to the early-time asymptote and the $\tilde{M}$ -solution to the large time asymptote of our problem. The main objective of this article is to accurately map the fracture evolution between the two asymptotes. This requires, among other things, to determine the regions of applicability of each asymptote, by conducting a perturbation analysis.

4 Tip Behavior

The behavior of the tip region of a fluid-driven crack is an important aspect of the problem under study. The tip asymptote not only yields important information about the propagation process but it can also be used to construct analytical models for simple geometries, or to provide the tip region boundary conditions for numerical models. Furthermore, as will be shown in this article, the analytical near-tip solution provides a surprisingly accurate solution for the whole fracture.

Before solving the problem posed in Sect. 2, it is thus necessary to first analyze in details, the nature of the solution near the crack tip, within the framework of the propagation regimes and scaling schemes already introduced. Such an analysis requires focusing on the tip region. This is simply achieved by introducing a new space variable $\hat{\xi}$ equal to the coordinate $(1 - \xi)$ divided by a small parameter, which is time-dependent and is thus necessarily related to the evolution parameter $\bar{C}_m$ in the M-scaling (and to $\bar{S}_m$ in the $\bar{M}$ -scaling). The stretching operation also effectively rejects the fracture inlet to infinity. In addition to the dilation of space, the fracture opening can also be rescaled so as to satisfy a time-independent equation governing $\hat{\Omega}(\hat{\xi})$, with a hat ($\hat{\cdot}$) denoting the autonomous tip solution. The time-independent solution can actually be interpreted as the solution for the problem of a semi-infinite hydraulic fracture propagating at constant speed (see Appendix B).

4.1 Tip Asymptotic Near the M-Vertex

It follows from the above discussion that the transformation of $(1 - \xi)$ and $\bar{\Omega}_m$ into the tip quantities $\hat{\xi}$ and $\hat{\Omega}$ is assumed to be of the form

$$(1 - \xi) = A_{m\hat{\xi}} \bar{C}_m^{a_{m\hat{\xi}}} \hat{\xi}, \quad \bar{\Omega}_m = A_{m\Omega} \bar{C}_m^{a_{m\Omega}} \hat{\Omega}, \quad (4.1)$$

where $A_{m\hat{\xi}}$, $a_{m\hat{\xi}}$, $A_{m\Omega}$, and $a_{m\Omega}$ are yet unknown constants. We also assume that $\hat{\Omega}$ is a power law of $\hat{\xi}$. In order to obtain these unknowns, we have to apply the above transformation to the governing Eq. 3.7, take the limit $\hat{\xi} \rightarrow 0$, integrate the result over $\hat{\xi}$, equate all powers of $\bar{C}_m$, and equate all numerical coefficients to unity. After proceeding through these steps, we obtain the expressions for the transformations

$$(1 - \xi) = \frac{1}{26} u_m^5 \bar{C}_m^{-6} \hat{\xi}, \quad \bar{\Omega}_m = \frac{1}{4} u_m^2 \bar{C}_m^{-2} \hat{\Omega}, \quad (4.2)$$

as well as the governing equation

$$\hat{\Omega}^3 \frac{d\hat{\Omega}}{d\hat{\xi}} = \hat{\Omega} + \hat{\xi}^{1/2}. \quad (4.3)$$

Here, $u_m = \dot{\ell}\tau/\ell = \dot{\gamma}_m\tau/\gamma_m + 4/5$ is a dimensionless tip velocity. The first and the second term of right-hand side of the governing Eq. 4.3 represent storage and leak-off effects, respectively.

A small $\hat{\xi}$ asymptotic expansion of $\hat{\Omega}$ can readily be deduced from (4.3)

$$\hat{\Omega}_0 = 3^{1/3} \hat{\xi}^{1/3} + \frac{2}{7} \hat{\xi}^{1/2} + O(\hat{\xi}^{2/3}), \quad \hat{\xi} \ll 1, \quad (4.4)$$

with the first term $\hat{\Omega}_0 = 3^{1/3} \hat{\xi}^{1/3}$ corresponding to the exact solution of (4.3) in the absence of the leak-off term $\hat{\xi}^{1/2}$. Hence, the small $\hat{\xi}$ asymptotic expansion of $\hat{\Omega}$ provides the tip behavior when the storage term dominates the leak-off one, i.e., when $\hat{\xi}^{1/2} \ll \hat{\Omega}$. It follows from (4.4) that the behavior of $\bar{\Omega}_m$ near the tip is

$$\bar{\Omega}_m \sim 3^{1/3} u_m^{1/3} (1 - \xi)^{1/3} + \frac{4}{7} \bar{C}_m u_m^{-1/2} (1 - \xi)^{1/2} + O(\bar{C}_m^2). \quad (4.5)$$

4.2 Tip Asymptotic Near the $\tilde{M}$ -Vertex

We proceed along similar lines to derive the tip asymptote near the $\tilde{M}$ -vertex. Transformation of the $\tilde{M}$ -scaled quantities into the tip quantities is given by

$$(1 - \xi) = \frac{1}{26} u_{\tilde{m}}^5 \bar{\mathcal{S}}_{\tilde{m}}^8 \hat{\xi}, \quad \bar{\Omega}_{\tilde{m}} = \frac{1}{4} u_{\tilde{m}}^2 \bar{\mathcal{S}}_{\tilde{m}}^3 \hat{\Omega}. \quad (4.6)$$

Here, the tip velocity is $u_{\tilde{m}} = \dot{\gamma}_{\tilde{m}} \tau / \gamma_{\tilde{m}} + 1/2$. This transformation yields the same governing Eq. 4.3, which can be solved for the large $\hat{\xi}$ expansion of $\hat{\Omega}$

$$\hat{\Omega}_{\infty} = \frac{2^{3/4}}{3^{1/4}} \hat{\xi}^{3/8} + \frac{2^{3/2} 3^{1/2}}{11} \hat{\xi}^{1/4} + O(\hat{\xi}^{1/8}) \quad \hat{\xi} \ll 1 \quad (4.7)$$

with the first term $\hat{\Omega}_{\infty} = (8/3)^{1/4} \hat{\xi}^{3/8}$ corresponding to the absence of the storage term $\hat{\Omega}$ in (4.3). Thus, the large $\hat{\xi}$ expansion of $\hat{\Omega}$ pertains to the tip behavior when the fracture propagation is dominated by leak-off. Hence, the behavior of $\bar{\Omega}_{\tilde{m}}$ near the tip is

$$\bar{\Omega}_{\tilde{m}} \sim \frac{2}{3^{1/4}} u_{\tilde{m}}^{1/8} (1 - \xi)^{3/8} + \frac{2 \times 3^{1/2}}{11} \bar{\mathcal{S}}_{\tilde{m}} u_{\tilde{m}}^{3/4} (1 - \xi)^{1/4} + O(\bar{\mathcal{S}}_{\tilde{m}}^2). \quad (4.8)$$

The PKN tip solution is thus characterized by the storage-dominated asymptote in the near-field (small $\hat{\xi}$) and by the leak-off-dominated asymptote in the far-field (large $\hat{\xi}$), in opposition to the spatial arrangement of the asymptotes near a fracture tip under plane strain conditions (in a plane perpendicular to the fracture edge) (Adachi and Detournay 2008; Mitchell et al. 2007a). In other words, there is always a small region at the tip of a PKN fracture, where the leak-off is negligible, even when the whole fracture is propagating in the leak-off-dominated $\tilde{M}$ -regime. In this asymptotic regime, the structure of the solution in the tip region is characterized by the existence of a boundary layer, which corresponds to the near-field storage-dominated tip asymptote. This unphysical situation is actually an artifact of the local elasticity assumption, which breaks down in the neighborhood of the tip over a distance comparable to the fracture height. It can be corrected, in principle, by recognizing that the relationship between the fracture aperture and the net pressure transitions from a non-local response at the tip to a local one away from the tip, as recently demonstrated in an elastic analysis of a planar fracture with a PKN geometry (Adachi and Peirce 2007).

4.3 Transient Tip Asymptotic

The first term of the near-field (small $\hat{\xi}$) and the far-field (large $\hat{\xi}$) tip asymptotes (respectively, $\hat{\Omega}_0$ and $\hat{\Omega}_{\infty}$) are actually very close to each other. Therefore, we are looking to construct a general solution of (4.3) of the form

$$\hat{\Omega} = \hat{\omega} \hat{\Omega}_0, \quad (4.9)$$

where $\hat{\omega}(\hat{\xi})$ is now the sought function. Its limit for $\hat{\xi} = 0$ and its behavior for large $\hat{\xi}$ are given by

$$\begin{aligned} \hat{\omega} &= 1, \quad \text{for } \hat{\xi} \rightarrow 0, \\ \hat{\omega} &= \hat{\Omega}_{\infty} / \hat{\Omega}_0 = 3^{-1/3} (8/3)^{1/4} \hat{\xi}^{1/24}, \quad \text{for } \hat{\xi} \rightarrow \infty. \end{aligned} \quad (4.10)$$

After substitution of (4.4) into (4.3), we obtain

$$(\hat{\omega}^4 - \hat{\omega}) \hat{\Omega}_0 + \hat{\Omega}_0^4 \hat{\omega}^3 \frac{d\hat{\omega}}{d\hat{\xi}} = \hat{\xi}^{1/2}. \quad (4.11)$$

It follows from (4.10) that

$$\frac{d \ln \hat{\omega}}{d \ln \hat{\xi}} < \frac{1}{24} \ll \frac{1}{3} < \frac{d \ln \hat{\Omega}}{d \ln \hat{\xi}}. \quad (4.12)$$

Therefore, we seek a solution in the form $\hat{\omega} = \hat{\omega}_0 + \hat{\omega}_1 + \dots$, where $\hat{\omega}_1 \ll \hat{\omega}_0$, and while we solve (4.11) for $\hat{\omega}_0$, we can neglect the derivative of $\hat{\omega}_0$. In other words, $\hat{\omega}_0$ and $\hat{\omega}_1$ are the solutions of the following algebraic equations

$$\hat{\omega}_0^4 - \hat{\omega}_0 = \hat{\Omega}_m^{-1} \hat{\xi}^{1/2} = 3^{-1/3} \hat{\xi}^{1/6}, \quad (4.13)$$

$$(4\hat{\omega}_0^3 - 1) \hat{\omega}_1 \hat{\Omega}_m + \hat{\Omega}_m^4 \hat{\omega}_0^3 \frac{d\hat{\omega}_0}{d\hat{\xi}} = 0. \quad (4.14)$$

The solution of (4.13) is given by

$$\hat{\omega}_0 = \frac{1}{2 \times 6^{1/3} \sqrt{m}} \left(m + \sqrt{12 - m^{3/2}} \right), \quad (4.15)$$

where $m = 2^{1/3}b - a/b$ with $a = 8\hat{\xi}^{1/6}$ and $b = \left(9 + \sqrt{81 + 256\hat{\xi}^{1/2}} \right)^{1/3}$. Substitution of (4.13) into (4.14) yields

$$\hat{\omega}_1 = -\frac{1}{2} \frac{\hat{\omega}_0^4 (\hat{\omega}_0^3 - 1)}{(4\hat{\omega}_0^3 - 1)^2}. \quad (4.16)$$

One can check that for large $\hat{\xi}$, $\hat{\omega}_0 = 3^{-1/12} \hat{\xi}^{1/24}$ differs from $\hat{\omega}$ less than on 3%, and $\hat{\omega}_0 + \hat{\omega}_1 = (31/32) 3^{-1/12} \hat{\xi}^{1/24}$ differs from $\hat{\omega}$ less than on 0.23%. The tip solution $\hat{\Omega}(\hat{\xi})$ is plotted in Fig. 3.

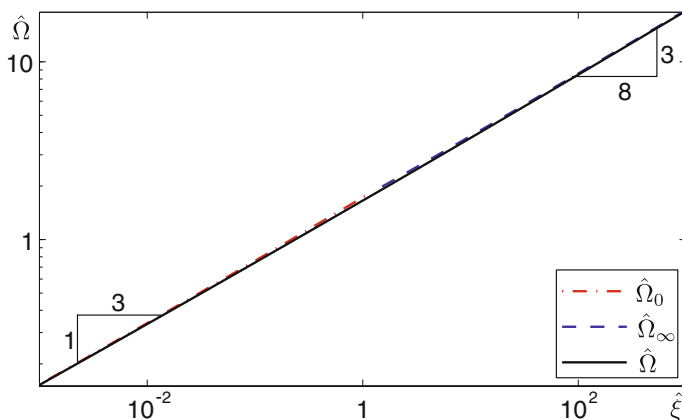


Fig. 3 Tip solution $\hat{\Omega}(\hat{\xi})$: near-field tip asymptote $\hat{\Omega}_0$ (dot-dashed line), far-field asymptote $\hat{\Omega}_\infty$ (dashed line), and transient solution $\hat{\Omega}$ (solid line)

5 Time-Asymptotic Solutions

In this section, we proceed to determine the small and large time asymptotic solutions.

5.1 M- and Near-M-Vertex Solution

Formulation of the mathematical model in the M-scaling has already indicated the existence of a similarity solution, identified as the M-solution, when $\bar{C}_m = 0$. Here, we seek not only to compute the M-solution but also the near-M-solution, the perturbation on the small time solution when $\bar{C}_m \ll 1$. Let $\bar{\mathcal{F}}_m(\xi, \bar{C}_m) = \{\gamma_m, \bar{\Omega}_m\}$ denotes the general solution in the M-scaling, and $\bar{\mathcal{F}}_{m0}(\xi) = \bar{\mathcal{F}}_m(\xi, 0)$ denotes the similarity solution at the M-vertex. For $\bar{C}_m \ll 1$, we seek an expansion of $\bar{\mathcal{F}}_m$ in the form

$$\bar{\mathcal{F}}_m = \bar{\mathcal{F}}_{m0} + \bar{C}_m \bar{\mathcal{F}}_{m1} + O(\bar{C}_m^2), \quad (5.1)$$

where $\bar{\mathcal{F}}_{m1} = \{\gamma_{m1}, \bar{\Omega}_{m1}\}$ is the first-order correction. After substituting this expansion into the Reynolds equation (3.7) and collecting the $O(1)$ and $O(\bar{C}_m)$ terms, we have

$$O(1): \quad \bar{\Omega}_{m0} - 4\xi \frac{d\bar{\Omega}_{m0}}{d\xi} = \frac{5}{4} \frac{d^2 \bar{\Omega}_{m0}^4}{d\xi^2}, \quad (5.2)$$

$$\begin{aligned} O(\bar{C}_m): \quad & \frac{1}{2} \bar{\Omega}_{m1} - \frac{4}{5} \xi \frac{d\bar{\Omega}_{m1}}{d\xi} + \frac{\gamma_{m1}}{\gamma_{m0}} \left(\frac{1}{5} \bar{\Omega}_{m0} - \frac{3}{10} \xi \frac{d\bar{\Omega}_{m0}}{d\xi} \right) + \frac{1}{\sqrt{1-\xi^{5/4}}} \\ & = \frac{d^2}{d\xi^2} (\bar{\Omega}_{m0}^3 \bar{\Omega}_{m1}). \end{aligned} \quad (5.3)$$

Here, we used the known power law $\ell \sim t^{4/5}$ at the M-vertex, so that $\theta = \xi^{5/4}$. Also, expanding the boundary condition (3.8), and collecting the $O(1)$ and $O(\bar{C}_m)$ terms gives

$$O(1): \quad \left. \frac{d\bar{\Omega}_{m0}^4}{d\xi} \right|_{\xi=0} = -2\gamma_{m0}^{-5/3}, \quad (5.4)$$

$$O(\bar{C}_m): \quad \left. \frac{d}{d\xi} (\bar{\Omega}_{m0}^3 \bar{\Omega}_{m1}) \right|_{\xi=0} = \frac{5}{6} \gamma_{m0}^{-8/3} \gamma_{m1}. \quad (5.5)$$

In order to find the solution $\bar{\mathcal{F}}_{m0}$, we have to solve the system of Eqs. 5.2 and 5.4. The solution is given by (see Appendix C)

$$\bar{\Omega}_{m0} = \left(\frac{12}{5} \right)^{1/3} (1-\xi)^{1/3} \left[1 - \frac{1-\xi}{96} + \frac{23(1-\xi)^2}{64512} - \frac{7(1-\xi)^3}{1327104} + \dots \right], \quad (5.6)$$

$$\gamma_{m0} = 0.660422. \quad (5.7)$$

This result agrees with the solution obtained by Kemp (1990). Note that the small time solution of Nordgren (1972), interpreted in the M-scaling corresponds to $\gamma_{m0} = 0.676$ and $\Omega_{m0}(0) = 0.99$ (in lieu of 1.00514 for this solution). Interestingly, we see that the M-solution (5.6) almost coincides with the tip asymptote (4.5), with an error of about 1%.

In order to find the correction $\bar{\mathcal{F}}_{m1}$, we have to solve system of Eqs. 5.3 and 5.5, see Appendix D for details. The region of applicability of the M-solution can be assessed from the correction to the M-solution, but alternatively from the tip solution (4.5), in view of

its dominance on the global solution. Using (4.5), in conjunction with, the global volume balance (3.6) yields $\gamma_{m1}/\gamma_{m0} \approx -1.03$. Therefore, the M-solution gives an accuracy κ for $\tau < 0.36\kappa^{10/3}$. For example, we have $\tau < 7.7 \times 10^{-8}$ for $\kappa = 0.01$.

5.2 $\tilde{M}$ - and Near- $\tilde{M}$ -Solutions

Here, we are looking for a large time expansion of the solution of the form

$$\tilde{\mathcal{F}}_{\tilde{m}} = \tilde{\mathcal{F}}_{\tilde{m}0} + \tilde{\mathcal{S}}_{\tilde{m}} \tilde{\mathcal{F}}_{\tilde{m}1} + O(\tilde{\mathcal{S}}_{\tilde{m}}^2), \quad (5.8)$$

where $\tilde{\mathcal{F}}_{\tilde{m}0} = \{\gamma_{\tilde{m}0}, \tilde{\Omega}_{\tilde{m}0}\}$ is the $\tilde{M}$ -solution and $\tilde{\mathcal{F}}_{\tilde{m}1} = \{\gamma_{\tilde{m}1}, \tilde{\Omega}_{\tilde{m}1}\}$ is the first-order correction. After substitution of this expansion into the global conservation law (3.9) and collecting the $O(1)$ and $O(\tilde{\mathcal{S}}_{\tilde{m}})$ terms, we have

$$O(1) : \frac{1}{2} = 2\gamma_{\tilde{m}0} \int_0^1 \sqrt{1 - \xi^2} d\xi, \quad (5.9)$$

$$O(\tilde{\mathcal{S}}_{\tilde{m}}) : 0 = \gamma_{\tilde{m}0} \int_0^1 \tilde{\Omega}_{\tilde{m}0} d\xi + 2\gamma_{\tilde{m}1} \int_0^1 \sqrt{1 - \xi^2} d\xi. \quad (5.10)$$

The Reynolds's Equation 3.10 yields

$$O(1) : \frac{1}{\sqrt{1 - \xi^2}} = \frac{1}{4} \frac{d^2 \tilde{\Omega}_{\tilde{m}0}^4}{d\xi^2}, \quad (5.11)$$

$$O(\tilde{\mathcal{S}}_{\tilde{m}}) : \frac{1}{8} \tilde{\Omega}_{\tilde{m}0} - \frac{1}{2} \xi \frac{\partial \tilde{\Omega}_{\tilde{m}0}}{\partial \xi} = \frac{\partial^2}{\partial \xi^2} (\tilde{\Omega}_{\tilde{m}0}^3 \tilde{\Omega}_{\tilde{m}1}). \quad (5.12)$$

Here, we have used the power law growth rule $\ell \sim t^{1/2}$ at the $\tilde{M}$ -vertex; so $\theta = \xi^2$. Equation (5.9) yields

$$\gamma_{\tilde{m}0} = \pi^{-1}, \quad (5.13)$$

and from (5.11), we can immediately find $\tilde{\Omega}_{\tilde{m}0}$

$$\tilde{\Omega}_{\tilde{m}0}(\xi) = (2\pi)^{1/4} \left(\frac{2}{\pi} \xi \arcsin \xi + \frac{2}{\pi} \sqrt{1 - \xi^2} - \xi \right)^{1/4}, \quad (5.14)$$

which is in accordance with Nordgren (1972). The series expansion of this solution is

$$\tilde{\Omega}_{\tilde{m}0}(\xi) = \frac{2^{7/8}}{3^{1/4}} (1 - \xi)^{3/8} \left[1 + \frac{1}{80} (1 - \xi) + \frac{159}{89600} (1 - \xi)^2 + \dots \right]. \quad (5.15)$$

The $\tilde{M}$ -solution matches the tip asymptotic (4.8) with an error of about 1.25%.

The correction to $\gamma_{\tilde{m}0}$ can be easily found from (5.10) by using (5.14)

$$\gamma_{\tilde{m}1}/\gamma_{\tilde{m}0} = -0.65, \quad (5.16)$$

while the correction $\tilde{\Omega}_{\tilde{m}1}$ can be calculated by double integration of (5.12). Although this correction can be easily found numerically, it does not appear possible to derive an analytical expression for it.

We can readily estimate the $\tilde{M}$ -region, the domain of applicability of the $\tilde{M}$ -solution, from (5.16). The $\tilde{M}$ -solution gives an accuracy κ for $\tau > 0.07\kappa^{-8/3}$. For example for $\kappa = 0.01$, we have $\tau > 1.5 \times 10^4$.

6 Transient Solution

6.1 Intermediate Scaling

The transient solution (referred to as the “MM̃-solution”) that connects the above asymptotes is computed numerically. The numerical method is based on an implicit finite volume algorithm formulated on a moving or stretching mesh. The growing fracture is discretized with a constant number of regularly spaced nodes. A detailed description of the algorithm can be found in Appendix E.

The algorithm is based on dimensionless equations expressed in the MM̃-scaling, which is intermediate between the M- and M̃-scaling. These two scalings cannot be used for the transient solution, as each one accentuates one of the asymptotic regimes and overshadows the other one. In the MM̃-scaling, the evolution parameter is τ , and the parameters $W_{m\tilde{m}}$ and $L_{m\tilde{m}}$ do not depend on time (see Appendix A for details)

$$W_{m\tilde{m}} = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^2}{C^{1/2} H} \right)^{1/3}, \quad L_{m\tilde{m}} = \left(\frac{\bar{\mu}}{\bar{E}} \frac{Q_0^5}{C^{1/8} H^4} \right)^{1/3}. \quad (6.1)$$

The equivalence between this new scaling and the already introduced M- and M̃-scaling is given by

$$\gamma = \gamma_m \tau^{4/5} = \gamma_{\tilde{m}} \tau^{1/2}, \quad (6.2)$$

$$\Omega = \Omega_m \tau^{1/5} = \Omega_{\tilde{m}} \tau^{1/8}. \quad (6.3)$$

In the MM̃-scaling, the governing equation (2.7) and the boundary condition (2.9) transform as (see Appendix A for details of the derivation)

$$\dot{\Omega} - \xi \frac{\dot{\gamma}}{\gamma} \frac{\partial \Omega}{\partial \xi} + \frac{\tau^{-1/2}}{\sqrt{1-\theta}} = \frac{1}{4\gamma^2} \frac{\partial^2 \Omega^4}{\partial \xi^2}, \quad (6.4)$$

$$\left. \frac{\partial \Omega^4}{\partial \xi} \right|_{\xi=0} = -2\gamma. \quad (6.5)$$

6.2 Results of Numerical Simulation

The numerical algorithm was implemented in *Mathematica* 5.2. Results of the simulation presented in Figs. 4, 5, 6, 7, and 8 (together with the asymptotic solutions expressed in the MM̃-scaling) were obtained with the following parameters: number of control volumes $n = 10$, initial time $\tau_0 = 10^{-8}$, and time step $\Delta\tau = 0.1\tau$.

The evolution of the crack length $\gamma(\tau)$ and inlet opening $\Omega(0, \tau)$ is shown in Figs. 4 and 5, respectively. The transient solution, which uses the small-time asymptote as an initial condition (at $\tau = 10^{-8}$), indeed converges to the large time asymptote. Table 1 tabulates the fracture length $\gamma(\tau)$, the tip velocity $u(\tau)$, and the opening at the inlet $\Omega(0, \tau)$ for selected values of time τ ranging on a logarithmic scale from 10^{-5} to 10^3 . Figures 6, 7, and 8 show the opening profile at early ($\tau = 10^{-3}$), intermediate ($\tau = 10^{-1}$), and large time ($\tau = 10^3$). We see that the early time solution is well-described by the M-asymptote and that there is excellent agreement between the large-time solution and the M̃-asymptote.

The tip solution remains very close to the numerical solution over the whole fracture at any time, so confirming the match between the global and tip solutions, that was observed at the M- and M̃-solution (with an accuracy greater than 2%).

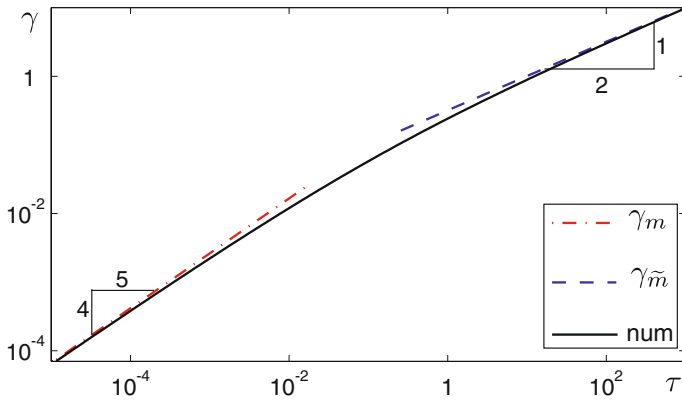


Fig. 4 Fracture half-length γ versus time τ (M̃M-scale): numerical results (solid line), γ_{m0} (left dot-dashed line), and $\gamma_{\tilde{m}0}$ (right dashed line)

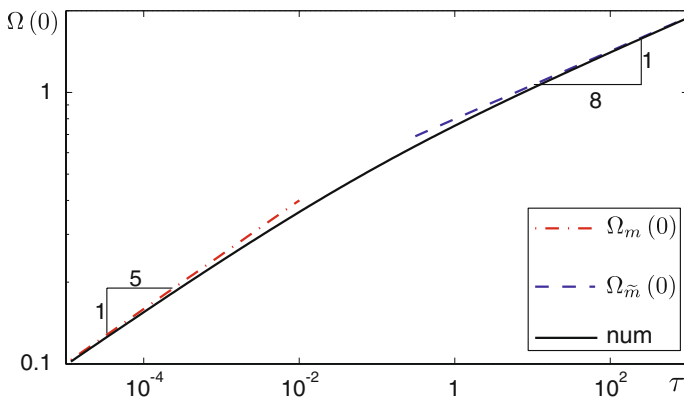


Fig. 5 Fracture opening at the inlet $\Omega(0)$ versus time τ (M̃M-scale): numerical results (solid line), Ω_{m0} (left dot-dashed line), and $\Omega_{\tilde{m}0}$ (right dashed line)

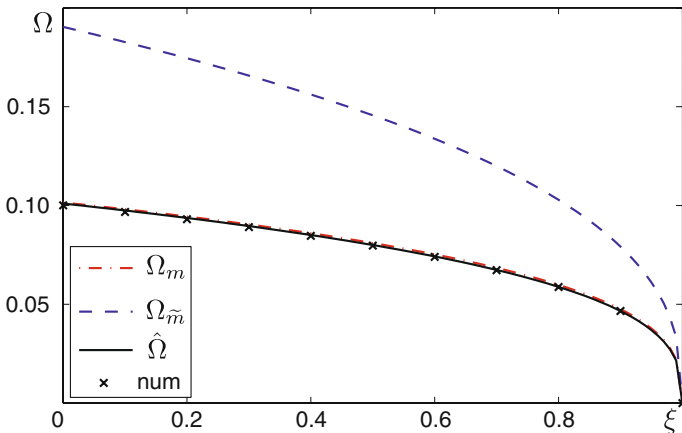


Fig. 6 Fracture opening Ω versus position ξ (M̃M-scale) at $\tau = 10^{-5}$: numerical results (marked by "x"), tip solution (solid line), Ω_{m0} (dot-dashed line), and $\Omega_{\tilde{m}0}$ (dashed line)

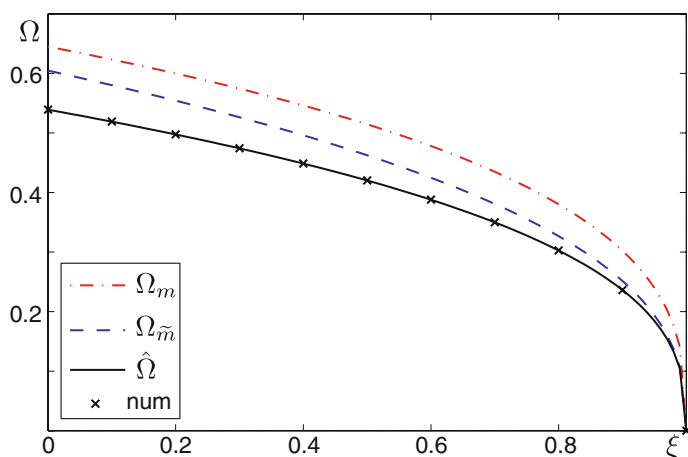


Fig. 7 Fracture opening Ω versus position ξ (MM-scale) at $\tau = 10^{-1}$: numerical results (marked by “x”), tip solution (solid line), Ω_{m0} (dot-dashed line), and $\Omega_{\tilde{m}0}$ (dashed line)

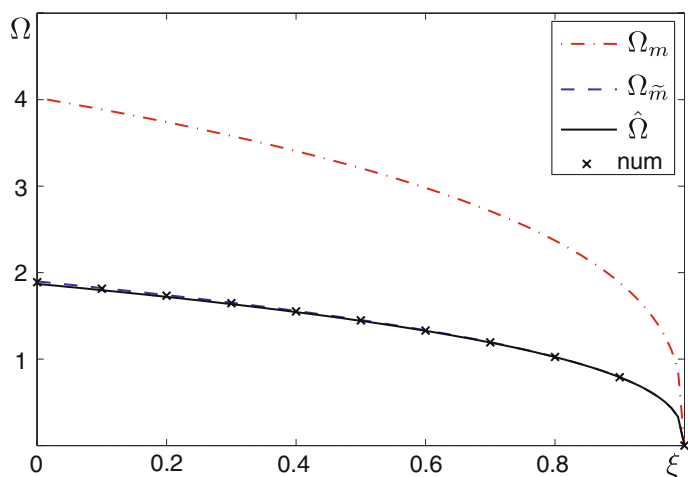


Fig. 8 Dimensionless fracture opening Ω versus position ξ (M̃M-scale) at $\tau = 10^3$: numerical results (marked by “x”), tip solution (solid line), Ω_{m0} (dot-dashed line), and $\Omega_{\tilde{m}0}$ (dashed line)

7 Conclusions

In this article, we have revisited the classical PKN model of a hydraulic fracture propagating in a permeable homogeneous elastic rock mass. Using scaling arguments, we have shown that the general dependence of the problem on time t and on five parameters, H , Q_0 , $\bar{E}$, $\bar{\mu}$, and C' , can be reduced to the dependence on, one only evolution parameter $\mathcal{C}_m$ (or $S_{\tilde{m}}$) or equivalently on a dimensionless time τ . Hence, a universal solution of this problem exists, as already noted by Nordgren (1972). From the scaling analysis, we also proved the existence of two similarity solutions (the M- and $\tilde{M}$ -solutions) at small and large times, which correspond, respectively, to storage- and leak-off-dominated regime.

Table 1 Numerical solution of the PKN fracture

τ	γ	u	$\Omega(0)$
1×10^{-5}	6.28976×10^{-5}	7.95259×10^{-1}	9.90971×10^{-2}
2×10^{-5}	1.08442×10^{-4}	7.92483×10^{-1}	1.13483×10^{-1}
5×10^{-5}	2.22002×10^{-4}	7.8785×10^{-1}	1.35596×10^{-1}
1×10^{-4}	3.80477×10^{-4}	7.83455×10^{-1}	1.5499×10^{-1}
2×10^{-4}	6.49881×10^{-4}	7.78124×10^{-1}	1.76988×10^{-1}
5×10^{-4}	1.30973×10^{-3}	7.69368×10^{-1}	2.10457×10^{-1}
1×10^{-3}	2.21221×10^{-3}	7.61201×10^{-1}	2.39519×10^{-1}
2×10^{-3}	3.71276×10^{-3}	7.51504×10^{-1}	2.7209×10^{-1}
5×10^{-3}	7.27311×10^{-3}	7.36087×10^{-1}	3.20895×10^{-1}
1×10^{-2}	1.19724×10^{-2}	7.223×10^{-1}	3.6256×10^{-1}
2×10^{-2}	1.94966×10^{-2}	7.06735×10^{-1}	4.0839×10^{-1}
5×10^{-2}	3.64884×10^{-2}	6.83702×10^{-1}	4.75784×10^{-1}
1×10^{-1}	5.77561×10^{-2}	6.64932×10^{-1}	5.32042×10^{-1}
2×10^{-1}	9.01811×10^{-2}	6.45679×10^{-1}	5.9292×10^{-1}
5×10^{-1}	1.59286×10^{-1}	6.20597×10^{-1}	6.81031×10^{-1}
1	2.41206×10^{-1}	6.02895×10^{-1}	7.53507×10^{-1}
2	3.60999×10^{-1}	5.86753×10^{-1}	8.31663×10^{-1}
5	6.05826×10^{-1}	5.68668×10^{-1}	9.44231×10^{-1}
1×10^1	8.87017×10^{-1}	5.5731×10^{-1}	1.03703
2×10^1	1.28973	5.47913×10^{-1}	1.13734
5×10^1	2.09688	5.3814×10^{-1}	1.28261
1×10^2	3.0128	5.3244×10^{-1}	1.40319
2×10^2	4.31451	5.27916×10^{-1}	1.53413
5×10^2	6.90506	5.23391×10^{-1}	1.72445

A lot of attention was devoted to the consideration of the tip region, motivated in part by the need to have a tip element for the computational mesh. We have derived not only the asymptotic tip behaviors near the near the M- and $\tilde{M}$ -vertex but also an approximate general asymptotic solution in the tip region. We have also shown that the tip solution matches the global M- and $\tilde{M}$ -solution with an error less than 2%. Numerical simulation has confirmed that the transient solution also can be approximated by the tip solution.

Further study on the PKN model should implement a tip asymptotic solution that does not rely on the local elasticity assumption, but that embeds instead a relationship with a built-in transition from a non-local plane strain condition—in a plane parallel to the bounding layers and perpendicular to the fracture, to a local plane strain condition—in a plane perpendicular to both the fracture and the bounding layers (Adachi and Peirce 2007). Such extension of the PKN model would allow to properly account for the rock toughness and would also overcome the issue regarding the improper spatial ordering of the tip asymptotes that was identified in this study.

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Appendix A: Scaling and Similarity Formulations

This appendix describes a reduction of the governing equations into their scaled form. Using (3.3), and making the transition from the spatial coordinate x to $\xi = x/\ell$, the global volume

balance equation 2.11 can be reduced to

$$\frac{Q_0 t}{2H} = \gamma L W \int_0^1 \Omega d\xi + 2C' \gamma L t^{1/2} \int_0^1 \sqrt{1 - \theta(\xi)} d\xi. \quad (\text{A.1})$$

At the same time Reynolds Equation 2.7 is transformed to

$$\dot{W}\Omega + W\dot{\Omega} - \xi W \left(\frac{\dot{\gamma}}{\gamma} + \frac{\dot{L}}{L} \right) \frac{\partial \Omega}{\partial \xi} + \frac{C'}{t^{1/2}} \frac{1}{\sqrt{1 - \theta}} = \frac{\bar{E}}{4H\bar{\mu}} \frac{W^4}{\gamma^2 L^2} \frac{\partial^2 \Omega^4}{\partial \xi^2}, \quad (\text{A.2})$$

and the boundary condition (2.9) to

$$\left. \frac{\partial \Omega^4}{\partial \xi} \right|_{\xi=0} = -2\gamma \mathcal{G}_\mu. \quad (\text{A.3})$$

Further re-arrangement of these equations allows us to write them in the following form

$$\frac{1}{2\gamma} = \mathcal{G}_v \int_0^1 \Omega d\xi + 2\mathcal{G}_c \int_0^1 \sqrt{1 - \theta(\xi)} d\xi, \quad (\text{A.4})$$

$$\mathcal{G}_v \left[\frac{\dot{W}t}{W} \Omega + \dot{\Omega}t - \xi \left(\frac{\dot{\gamma}t}{\gamma} + \frac{\dot{L}t}{L} \right) \frac{\partial \Omega}{\partial \xi} \right] + \frac{\mathcal{G}_c}{\sqrt{1 - \theta}} = \frac{1}{\mathcal{G}_\mu} \frac{1}{4\gamma^2} \frac{\partial^2 \Omega^4}{\partial \xi^2}, \quad (\text{A.5})$$

where three-dimensionless groups have been identified

$$\mathcal{G}_v = \frac{LWH}{Q_0 t}, \quad \mathcal{G}_c = \frac{C'HL}{Q_0 t^{1/2}}, \quad \mathcal{G}_\mu = \frac{\bar{\mu}LQ_0}{\bar{E}W^4}. \quad (\text{A.6})$$

Notice that the quantities W and L are still to be determined.

In this particular problem, the only dissipation mechanism is viscous flow. This means, that we can set the viscosity-related group $\mathcal{G}_\mu$ equal to unity, $\mathcal{G}_\mu \equiv 1$. The other two-dimensionless groups, $\mathcal{G}_v$ and $\mathcal{G}_c$, are associated with the fluid volume components: storage in the fracture and leak-off to the formation, respectively. Hence, the proper scaling for each of the propagation regimes described in Sect. 3.1 (the storage-viscosity or M-regime and the leak-off-viscosity or $\tilde{M}$ regime) can be obtained by setting the appropriate dimensionless group equal to unity.

In this way, the parameters for the M-scaling, cf. (3.4), are obtained by taking $\mathcal{G}_\mu \equiv 1$ and $\mathcal{G}_v \equiv 1$, and parameter $\mathcal{C}_m$ is equal to $\mathcal{G}_c$, $\mathcal{C}_m \equiv \mathcal{G}_c$. In order to obtain the $\tilde{M}$ scaling, cf. (3.5), we set $\mathcal{G}_\mu \equiv 1$, $\mathcal{G}_c \equiv 1$, and $\mathcal{S}_m \equiv \mathcal{G}_v$. In order to obtain the $M\tilde{M}$ -scaling, cf. (6.1), we move all dependence on time to the left-hand sides of (A.6), by forcing the parameters W and L to be time-independent; i.e., we set $\mathcal{G}_\mu \equiv 1$, $\mathcal{G}_v \equiv \tau^{-1}$, and $\mathcal{G}_c \equiv \tau^{-1/2}$ in (A.6), where $\tau = t/t_{m\tilde{m}}$. Solving the resulting equations yields the expressions for the scale $t_{m\tilde{m}}$ in (3.2) and the scales $W_{m\tilde{m}}$ and $L_{m\tilde{m}}$ in (6.1).

Appendix B: Tip Region

B.1 Governing Equation

Analysis of the near-tip processes of a fluid-driven fracture can be performed by considering the problem of a semi-infinite fluid-driven fracture propagating with constant velocity V in

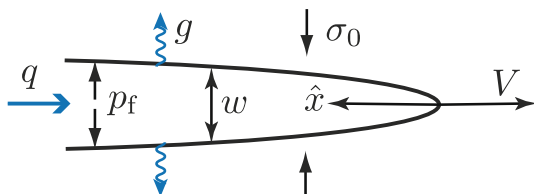


Fig. 9 Semi-infinite fluid-driven fracture

an elastic, permeable solid. Let us consider the moving coordinate axis $\hat{x}$, with origin at the tip, and pointing towards the fracture, see Fig. 9

$$\hat{x} = -x + \int_0^t V(\tau) d\tau = -x + \ell(t). \quad (\text{B.1})$$

In the moving coordinate, the continuity Eq. 2.2 combined with Poiseuille law (2.4) becomes

$$\frac{\partial \hat{w}}{\partial t} + V \frac{\partial \hat{w}}{\partial \hat{x}} - \frac{\bar{E}}{\bar{\mu} H} \frac{\partial}{\partial \hat{x}} \left(\hat{w}^3 \frac{\partial \hat{w}}{\partial \hat{x}} \right) + \hat{g} = 0, \quad (\text{B.2})$$

where the symbol $(\hat{\cdot})$ indicates that the variables are now functions of the moving coordinate $\hat{x}$.

Assuming that in the tip region $\hat{w} \sim \hat{x}^\alpha$ with $\alpha < 1$ (to be verified posteriori), $\partial \hat{w} / \partial t \ll V \partial \hat{w} / \partial \hat{x}$ as $\hat{x} \rightarrow 0$. The expression for the fluid loss velocity $\hat{g}$ is deduced by substituting the first term of the Taylor expansion of $t_0(x)$, $t_0 = t - \hat{x} / V$ into the leak-off Eq. 2.6

$$\hat{g} = C' (V / \hat{x})^{1/2}. \quad (\text{B.3})$$

The boundary condition at the tip is

$$\hat{w} = 0, \quad \text{at} \quad \hat{x} = 0. \quad (\text{B.4})$$

Substitution of Eq. B.3 into Eq. B.2 and integration over $\hat{x}$ yields

$$V \frac{d\hat{w}}{d\hat{x}} - \frac{\bar{E}}{\bar{\mu} H} \frac{d}{d\hat{x}} \left(\hat{w}^3 \frac{d\hat{w}}{d\hat{x}} \right) + C' \left(\frac{V}{\hat{x}} \right)^{1/2} = 0. \quad (\text{B.5})$$

While no constraint on the variation of V with t has been imposed to derive the above equation, we note that the same equation also governs the problem of a PKN semi-infinite hydraulic fracture propagating with constant velocity V . In its application to the asymptotic tip solution of a finite fracture, V has to be interpreted as the current tip velocity, however.

B.2 Scaling

We introduce a dimensionless coordinate $\hat{\xi}$ and crack opening $\hat{\Omega}$ as follows

$$\hat{x} = l_{m\tilde{m}} \hat{\xi}, \quad \hat{w} = w_{m\tilde{m}} \hat{\Omega}, \quad (\text{B.6})$$

where

$$l_{m\tilde{m}} = \frac{\bar{\mu}^2 H^2 V^5}{64 C'^6 \bar{E}^2}, \quad w_{m\tilde{m}} = \frac{\bar{\mu} H V^2}{4 C'^2 \bar{E}}. \quad (\text{B.7})$$

Scaling (B.5), according to (B.6), yields the following governing equation for the crack opening $\hat{\Omega}$

$$\hat{\Omega}^3 \frac{d\hat{\Omega}}{d\hat{\xi}} = \hat{\Omega} + \hat{\xi}^{1/2}. \quad (\text{B.8})$$

Appendix C: M-Solution

As was done in Kemp (1990), we are looking for a solution of (5.2) in the form

$$\bar{\Omega}_{m0}(\xi) = (1 - \xi)^{\alpha_m/4} [a_0 + a_1(1 - \xi) + a_2(1 - \xi)^2 + \dots]. \quad (\text{C.1})$$

From (C.1), we deduce

$$\bar{\Omega}_{m0}^4(\xi) = (1 - \xi)^{\alpha_m} [b_0 + b_1(1 - \xi) + b_2(1 - \xi)^2 + \dots]. \quad (\text{C.2})$$

The relationship between a_n and b_n can be obtained from the formula for the n -th derivative of the composition $f[g(x)]$ (Faà di Bruno's formula 19)

$$a_n = \sum_{k_1+2k_2+\dots+nk_n=n} \frac{b_1^{k_1} \dots b_n^{k_n}}{k_1! \dots k_n!} \frac{1}{4} \left(\frac{1}{4} - 1\right) \dots \left(\frac{5}{4} - k\right) b_0^{\frac{1}{4}-k}, \quad k = \sum_n k_n. \quad (\text{C.3})$$

Substitution of (C.2) into (5.4) leads to

$$\sum_n (\alpha_m + n) b_n = 2\gamma_m^{-5/3}. \quad (\text{C.4})$$

Substitution of (C.1) and (C.2) into (5.2) and equating like powers lead to

$$\alpha_m = 4/3, \quad a_0 = \left[\frac{5}{4}(\alpha_m - 1)\right]^{-1/3}, \quad b_0 = \left[\frac{5}{4}(\alpha_m - 1)\right]^{-4/3}, \quad (\text{C.5})$$

and for $n > 0$

$$(\alpha_m + 4n)a_n + (5 - \alpha_m - 4n)a_{n-1} = \frac{5}{4} [\alpha_m(\alpha_m - 1) + 2n\alpha_m + n(n - 1)] b_n. \quad (\text{C.6})$$

Formula (C.7) can be rewritten as follows

$$a_n = a_n^{n-1} + a_n^n = \sum_{\substack{k_1+2k_2+\dots+(n-1)k_{n-1}=n}} \frac{b_1^{k_1} \dots b_{n-1}^{k_{n-1}}}{k_1! \dots k_{n-1}!} \frac{1}{4} \left(\frac{1}{4} - 1\right) \dots \left(\frac{5}{4} - k\right) b_0^{\frac{1}{4}-k} + \frac{1}{4} b_0^{-3/4} b_n. \quad (\text{C.7})$$

By using the above Eq. C.6 yields

$$b_n = \frac{(\alpha_m + 4n)a_n^{n-1} + (5 - \alpha_m - 4n)a_{n-1}}{\frac{5}{4} [\alpha_m(\alpha_m - 1) + 2n\alpha_m + n(n - 1)] - \frac{1}{4}(\alpha_m + 4n)b_0^{-3/4}}. \quad (\text{C.8})$$

This is recurrence formula, i.e., if we know all a_i, b_i for $i < n$, we can calculate a_n and b_n . The coefficients a_0 and b_0 are given by (C.5). After we have determined all the coefficients a_i and b_i , we can calculate γ_{m0} using (C.4).

Appendix D: Correction to M-Solution

The near-M vertex solution has to have the tip asymptotic (4.5)

$$\bar{\Omega}_m \sim \left(\frac{12}{5}\right)^{1/3} \left(1 + \frac{1}{8} \bar{C}_m \frac{\gamma_{m1}}{\gamma_{m0}}\right) (1 - \xi)^{1/3} + \frac{2\sqrt{5}}{7} \bar{C}_m (1 - \xi)^{1/2} + O(\bar{C}_m^2). \quad (D.1)$$

Here, we took into account the dependence of the tip velocity u on time. It is reasonable to look for solution of (5.3) in the form

$$\begin{aligned} \bar{\Omega}_{m1} &= \frac{1}{8} \frac{\gamma_{m1}}{\gamma_{m0}} \bar{\Omega}_{m0} + \bar{\Omega}_{m1}^{(\gamma_{m0})} = \frac{1}{8} \frac{\gamma_{m1}}{\gamma_{m0}} \bar{\Omega}_{m0} + \frac{2\sqrt{5}}{7} (1 - \xi)^{1/2} \left[1 + \sum_{i=1}^{\infty} a_i (1 - \xi)^i \right] \\ &\quad + (1 - \xi)^{1/3} \sum_{i=1}^{\infty} b_i (1 - \xi)^i. \end{aligned} \quad (D.2)$$

After substitution of this equation into (5.3) and (5.5), we find

$$\frac{1}{2} \bar{\Omega}_{m1}^{(\gamma_{m0})} - \frac{4}{5} \xi \frac{d\bar{\Omega}_{m1}^{(\gamma_{m0})}}{d\xi} + \frac{13}{80} \frac{\gamma_{m1}}{\gamma_{m0}} \bar{\Omega}_{m0} + \frac{1}{\sqrt{1 - \xi^{5/4}}} = \frac{d^2}{d\xi^2} \left(\bar{\Omega}_{m0}^3 \bar{\Omega}_{m1}^{(\gamma_{m0})} \right), \quad (D.3)$$

$$\frac{d}{d\xi} \left(\bar{\Omega}_{m0}^3 \bar{\Omega}_{m1}^{(\gamma_{m0})} \right) \Big|_{\xi=0} = \frac{13}{12} \gamma_{m0}^{-8/3} \gamma_{m1}. \quad (D.4)$$

The coefficients a_i and b_i can be found numerically by minimization procedure. However, an excellent approximation of $\bar{\Omega}_{m1}$ is already obtained by neglecting all the terms with a_i and b_i .

Appendix E: Numerical Algorithm for the Transient Solution

This appendix describes the numerical algorithm used to solve the governing Eq. 6.4. The algorithm is based on an implicit finite volume scheme in a moving spacial grid. The proposed algorithm assumes that the solution of a semi-infinite crack propagating in a permeable medium can be used to approximate the asymptotic behavior of the field quantities in the near-tip region.

E.1 Description of Numerical Scheme

The crack is divided into n control volumes of constant length $\Delta\xi = 1/n$ (see Fig. 10). The time τ is discretized with a variable time step $\Delta\tau$, which is a monotonic function of τ . The numerical scheme uses a discretized form of the conservation law, i.e., the conservation law is enforced for each control volume. To keep the physics transparent, we start from the dimensional form of the conservation Eq. 2.2 and then rewrite it in dimensionless form.

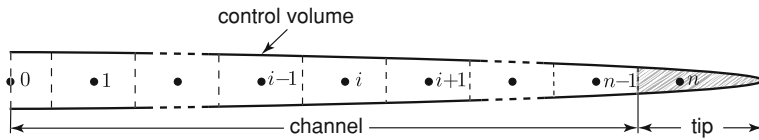


Fig. 10 Computational mesh

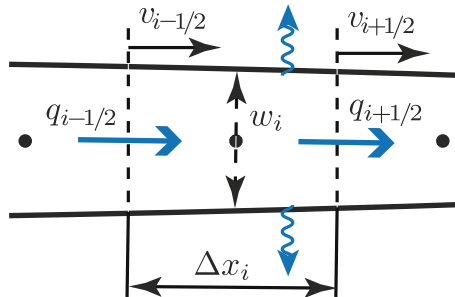


Fig. 11 Finite volume cell in the channel

E.1.1 Discrete Governing Equation in the Channel

The fluid volume balance in a control volume is given by, see Fig. 11

$$\Delta t [q_{i-1/2} - q_{i+1/2} - g_i \Delta x_i - v_{i-1/2} w_{i-1/2} + v_{i+1/2} w_{i+1/2}] = w_i \Delta x_i - w_i^{\text{old}} \Delta x_i^{\text{old}}. \quad (\text{E.1})$$

Here, $w_{i-1/2} = (w_{i-1} + w_i) / 2$, and for $q_{i-1/2}$ Eq. 2.5 yields

$$q_{i-1} = -\frac{\bar{E}}{4\bar{\mu}H} \frac{w_i^4 - w_{i-1}^4}{x_i - x_{i-1}}. \quad (\text{E.2})$$

Using the continuity relation, $\Delta x_i - \Delta x_i^{\text{old}} = (v_{i+1/2} - v_{i-1/2})\Delta t$ and transforming to the dimensionless variables, we find that

$$\begin{aligned} & \frac{\gamma^{\text{old}}}{\gamma} \frac{\Omega_i - \Omega_i^{\text{old}}}{\Delta \tau} - \frac{u}{\tau} \frac{\xi_{i+1/2} \Omega_{i+1} - \Delta \xi \Omega_i - \xi_{i-1/2} \Omega_{i-1}}{2\Delta \xi} + \frac{\tau^{-1/2}}{\sqrt{1-\theta}(\xi_i)} \\ &= \frac{1}{4\gamma^2} \frac{\Omega_{i+1}^4 - 2\Omega_i^4 + \Omega_{i-1}^4}{\Delta \xi^2}, \end{aligned} \quad (\text{E.3})$$

where $u = \dot{\gamma} \tau / \gamma$.

Following the same procedure, we obtain the discrete equation for the first control element

$$\begin{aligned} & \frac{3}{4} \frac{\gamma^{\text{old}}}{\gamma} \frac{3\Omega_0 + \Omega_1 - 3\Omega_0^{\text{old}}}{\Delta \tau} - \Omega_1^{\text{old}} - \frac{u}{\tau} \frac{\Omega_1 - \Omega_0}{4} + \frac{\tau^{-1/2}}{\sqrt{1-\theta}(\Delta \xi/4)} \\ &= \frac{1}{2\gamma^2} \frac{2\gamma \Delta \xi + \Omega_1^4 - \Omega_0^4}{\Delta \xi^2}. \end{aligned} \quad (\text{E.4})$$

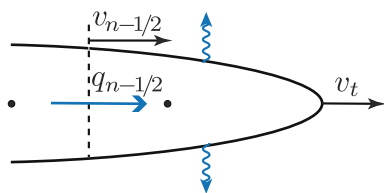


Fig. 12 Tip finite volume cell

E.1.2 Discrete Governing Equation in the Tip Region

The flow balance in this control volume is, see Fig. 12

$$\left[q_{n-1/2} - v_{n-1/2} w_{n-1/2} - \int_{x_{n-1/2}}^{x_t} g(x) dx \right] = \int_{x_{n-1/2}}^{x_t} w(x) dx - \int_{x_{n-1/2}}^{x_t} w(x) dx \Big|_{\text{old}}. \quad (\text{E.5})$$

The right-hand side can be transformed as follows

$$\begin{aligned} & \int_{x_{n-1/2}}^{x_t} w(x) dx - \int_{x_{n-1/2}}^{x_t} w(x) dx \Big|_{\text{old}} \\ &= \int_0^{x_t - x_{n-1/2}} w(x_t - \tilde{x}) d\tilde{x} - \int_0^{x_t - x_{n-1/2}} w(x_t - \tilde{x}) d\tilde{x} \Big|_{\text{old}} \\ &= \int_{x_t^{\text{old}} - x_{n-1/2}^{\text{old}}}^{x_t - x_{n-1/2}} w(x_t - \tilde{x}) d\tilde{x} + \int_0^{x_t^{\text{old}} - x_{n-1/2}^{\text{old}}} [w(x_t - \tilde{x}) - w^{\text{old}}(x_t - \tilde{x})] d\tilde{x}. \end{aligned}$$

Here, the first term can be rewritten

$$\int_{x_t^{\text{old}} - x_{n-1/2}^{\text{old}}}^{x_t - x_{n-1/2}} w(x_t - \tilde{x}) d\tilde{x} \approx w_{n-1/2} (v_t - v_{n-1/2}) \Delta t. \quad (\text{E.6})$$

We will omit the second term in the right-hand side, although this term could potentially be of the same order as the first one. This term is dropped in order to match the analytical tip solution, which is valid under the assumption of self-similarity ($\partial/\partial t \ll V \partial/\partial x \rightarrow w = w^{\text{old}}$).

Using (E.6) and transforming to dimensionless variables, (E.5) yields

$$\frac{u}{\tau} \frac{\Omega_n + \Omega_{n-1}}{2} + 2\sqrt{\frac{u}{\tau}} \sqrt{1 - \xi_{n-1/2}} = -\frac{1}{4\gamma^2} \frac{\Omega_n^4 - \Omega_{n-1}^4}{\Delta \xi}. \quad (\text{E.7})$$

This equation is used to update the velocity u .

E.2 Algorithm of Computation

The calculations are started from the small time asymptote. The sequence of computations for any time step is given by

- using the updated tip velocity u , calculate the crack length $\gamma = \gamma^{\text{old}} (1 - u\Delta\tau/\tau)^{-1}$,
- using the tip solution, the updated length γ , and tip velocity u calculate Ω_n ,
- solve the system of equation (E.3), (E.4) for Ω_i (Newton method),
- knowing the solution in the channel Ω_i using (E.7) compute the tip velocity u , and
- repeat until the solution has converged.

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